Result Table: Static vs Non-Static Counter in Multithreading

Program: jubayer_thread.java | Name: _____ | ID: _____

Experiment B: Unsynchronized static counter (mode = false)

Threads	Run	Expected	Static (unsafe)	Non-static total	Abs. diff	Diff (%)
1	1	1,000	1,000	1,000	0	0.0000%
2	1	20,000	10,750	20,000	9,250	46.2500%
2	2	20,000	13,042	20,000	6,958	34.7900%
5	1	50,000	13,190	50,000	36,810	73.6200%
10	1	500,000	127,602	500,000	372,398	74.4796%
100	1	5,000,000	4,155,270	5,000,000	844,730	16.8946%
100	2	5,000,000	3,734,523	5,000,000	1,265,477	25.3095%

Percentage difference = |static - non-static| / non-static x 100

Experiment A: Thread-safe static counter, AtomicLong (mode = true)

Threads	Expected	Static (AtomicLong)	Non-static total	Abs. diff	Diff (%)
1	1,000	1,000	1,000	0	0%
2	20,000	20,000	20,000	0	0%
5	50,000	50,000	50,000	0	0%